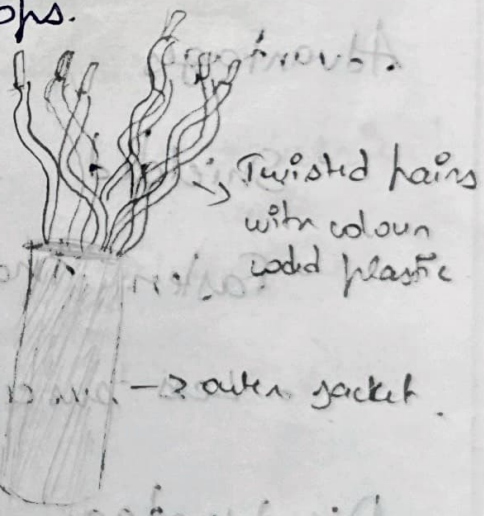
Experiment - 2 Date : 30/07/2026

Aim : Study of different types of Network cables

(i) Unshielded Twisted Pair (UTP) Cable

Category	Max data transmission
3	10 bps
5	Up to 100mbps
5e	1 Gbps.



Advantages

- Cheaper
- Easy to install

Disadvantages

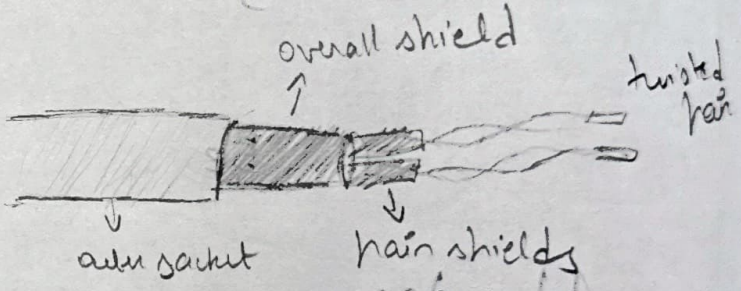
- More prone to Electromagnetic interference and noise.

Application

- Fast ethernet
- Gigabit ethernet

(ii) Shielded Twisted pair (STP)

Category	Max data transmission
STP	6, 6a	10 Gbps
SSTP	7	10Gbps



Advantages

- Shielded
- Faster than UTP
- Less susceptible to noise

Disadvantages

- Expensive
- Greater installation effort

Application

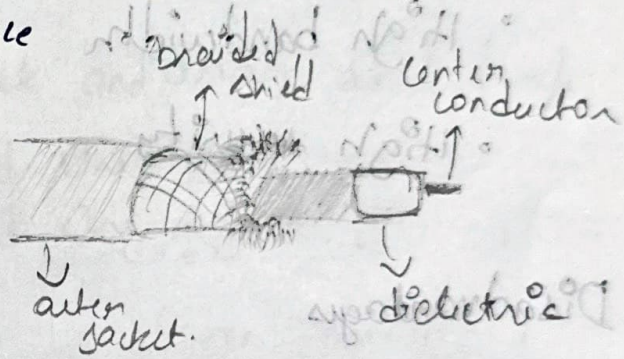
- Gigabit Ethernet
- 10G Ethernet

(iii) Coaxial Cable

RG6
RG 59
RG-11
} 10-100 Mbps max data transmission

Advantages

- High bandwidth
- Immune to interference
- Versatile



Disadvantages

- Limited distance
- Cost
- Size is bulky

Application

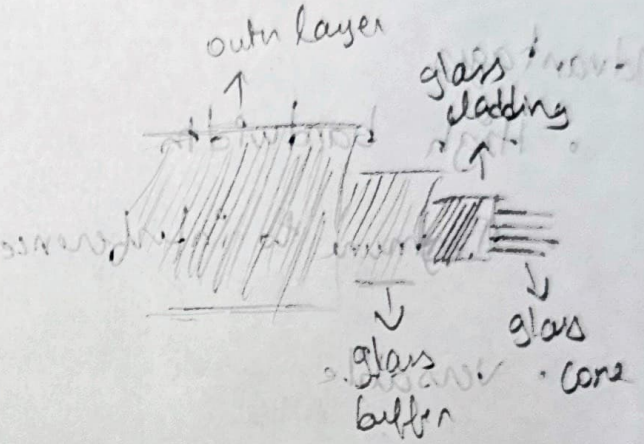
- Speed of signal is 500m
- Television network
- High speed internet.

(iv) Fibre optics cable

Single mode } 100 Gbps.
Multimode }

Advantages

- High speed
- High bandwidth
- High security



Disadvantages

- Expensive
- Required skill installers

Result - Successfully analysed the study of different types of Network Cables.